Dynamic Scheduling with Random Start and End Times

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Abstract

We study sequential interval scheduling when start and end times are random. The set of tasks and their weights are known in advance, while each task's start and end times are independent draws from known discrete distributions and are revealed only upon commitment. Conflicts with already committed tasks are resolved as intervals realize, and remaining tasks update their distributions accordingly. The objective is to maximize the expected weight of a conflict-free schedule. We propose two models that differ in how conflicts are enforced, develop LP relaxations and bounds for each, and present an SDP relaxation together with computational evidence.

1 Introduction

The *interval scheduling problem* is classical in operations research and computer science. It considers a set of tasks $N := [n]$ and consecutive time slots $M := [m]$. Each task is represented by an interval $[s_i, e_i]$ describing when it must be processed on a single resource. The *maximum interval scheduling problem* seeks a set of pairwise nonoverlapping intervals (a schedule) of maximum total weight. When all tasks have the same weight, the greedy rule that repeatedly selects the earliest-ending task and discards conflicts is optimal. The weighted version is also solvable in $\Theta(n)$ time [5].

An *interval graph* is an undirected graph representing intervals on a line, where each vertex corresponds to an interval and an edge connects two vertices if their intervals overlap. Thus a feasible schedule corresponds exactly to an independent set of the associated interval graph. Since all interval graphs are perfect, the maximum interval scheduling problem can be represented by the clique LP pair of the interval graph $G = (N, E)$,

$$\begin{aligned} \max_{x \geq 0} \quad & w^\top x \\ \text{s.t.} \quad & \sum_{i \in C} x_i \leq 1, \forall C \in \mathscr{C}(G) \end{aligned} \tag{LP-P}$$

$$\begin{aligned} \min_{\mu \geq 0} \quad & \sum_{C \in \mathscr{C}(G)} \mu_C \\ \text{s.t.} \quad & \sum_{C \ni i} \mu_C \geq w_i, \ \forall i \in N, \end{aligned} \tag{LP-D}$$

where $\mathscr{C}(G)$ is the set of all maximal cliques of G. (LP-P) formulates the maximum independent set problem and (LP-D) formulates the minimum cover problem. For perfect graphs, (LP-P) and (LP-D) are tight relaxations but not guaranteed to be solvable in polynomial time because there are possibly exponentially many maximal cliques. For an interval graph, the number of maximal cliques is at most $\min\{m, n\}$. Thus the deterministic maximum interval scheduling problem can be solved efficiently either directly or via these LP relaxations. Our goal is to extend this setting to incorporate uncertainty and sequential decisions while retaining tractable relaxations and algorithms.

Several stochastic interval scheduling models have been studied. A classical one is *online interval scheduling* [7], where intervals of fixed length are presented to the scheduler in the order of their start times and must be accepted or rejected irrevocably. The objective is to maximize total weight while maintaining feasibility. Recent work includes online interval scheduling with predictions of upcoming intervals [2] and models that allow overlaps but maximize a satisfaction objective [6]. Other variants include interval scheduling with random delays [3] and settings where tasks may depart unexpectedly and service times are stochastic [9].

We study the setting where the set of tasks and their weights are known in advance but their start and end times are random. The scheduler acts sequentially, learns realized intervals as tasks are committed, and seeks to maximize the expected total weight.

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2 Dynamic Scheduling with Random Start and End Times

We introduce the following model. Let $M = [m]$ be the set of slots, $N := [n]$ the set of tasks, and $w \in \mathbb{R}_+^N$ the task weights. For each task $i \in N$, let D_i^s, D_i^e be the discrete distributions of start and end times, and denote $S_i, E_i \subseteq M$ as their supports. We assume $\max S_i \leq \min E_i$ and define $D := \{D_i^* : * \in \{s, e\}, i \in N\}$. When task i is scheduled, its start and end times s_i, e_i are realized. For each remaining task j, we compute the probability p_{ij} that it conflicts with i based on D_j^s, D_j^e and draw $B_{ij} \sim \text{Bernoulli}(p_{ij})$ independently. If $B_{ij} = 1$, task j is deleted; otherwise it remains available and its distributions are updated to condition on the slots occupied by the committed task. Our goal is to schedule tasks sequentially and maximize the expected total weight of scheduled tasks. We call this problem *dynamic scheduling with random start and end times (DSRSE)*.

DSRSE can be formulated as a dynamic program with Bellman recursion,

$$\max_{\pi \in \Pi} \mathbb{E}_\pi [V^\pi(N, M, w, D)]; \qquad V^*(N, M, w, D) := \max_i \{w_i + \mathbb{E}[V^*(N', M', w, D')]\},$$

where Π is the set of all policies and $V^\pi(N, M, w, D)$ is a random variable representing the total weight of scheduled tasks under policy $\pi \in \Pi$. V^* is the optimal value of the dynamic programming problem, and N', M', w, D' are the random variables representing the resulting state after scheduling task i. Like many dynamic programming problems, this problem has exponentially many states and each state has $O(n)$ actions, so it is unreasonable to solve it directly. In general, this problem is NP-hard.

Proposition 2.1. *DSRSE is NP-hard.*

Proof. Suppose there are $n+1$ tasks and $2n$ slots. Let 0 be a task taking positions $1, \ldots, n$ with probability 1, and task i taking position i with probability p_i and $n+i$ with probability $(1-p_i)$. Then this problem is equivalent to the dynamic maximum stable set problem over a star graph, where 0 is the center. This problem is NP-hard [8]. $\square$

Example 2.2. *Consider an instance with $N = [2], M = [3]$, and unit weights, where $D_1^s = \mathbb{1}_{\{1\}}, D_1^e = \mathbb{1}_{\{2\}}$, $D_2^s = U(\{2,3\}), D_2^e = \mathbb{1}_{\{3\}}$. If the scheduler commits to task 1 first, then task 1 occupies $[1,2]$.*

For DSRSE, task 2 has probability $1/2$ to be deleted and $1/2$ to remain. Flip a fair coin. If it remains, task 2's distributions are updated as $D_2^s = \mathbb{1}_{\{3\}}, D_2^e = \mathbb{1}_{\{3\}}$. On the other hand, if task 2 is scheduled first, then if it is realized as $[2,3]$, task 1 is deleted; if it is realized as $[3]$, task 1 remains.

When s_i, e_i are deterministic for every task, DSRSE reduces to the maximum independent set problem of the corresponding interval graph. As discussed above, it can be represented by (LP-P) and (LP-D), so the deterministic problem can be solved efficiently via these LP relaxations. For an interval graph, all maximal cliques correspond to the slots M, so the constraint $\sum_{i \in C} x_i \leq 1, \forall C \in \mathscr{C}(G)$ in (LP-P) simply enforces that each slot is occupied by at most one task. We consider analogous constraints for DSRSE.

2.1 Dynamic Formulation and Relaxations

For DSRSE, consider $P^s, P^e, P^o \in \mathbb{R}^{n \times m}$, where $P_{ik}^s, P_{ik}^e, P_{ik}^o$ are the probabilities that task i starts at, ends at, and occupies slot k, respectively. Let $T = [\min\{m, n\}]$ be the stages of the dynamic program. For a policy π, let $x_{ik}^{ts}, x_{ik}^{te}, x_{ik}^{to}$ be the probabilities that i is scheduled and starts at, ends at, or occupies slot k at stage t, respectively. Let $x_i^t = \sum_{k=1}^m x_{ik}^{ts} = \sum_{k=1}^m x_{ik}^{te}$ be the probability of scheduling i at stage t. The objective can then be written as

$$\max_{\pi \in \Pi} \sum_t \sum_i w_i x_i^t.$$

Analogous to the deterministic case, we impose that the probability a slot is occupied is at most 1. We relax DSRSE to

$$\begin{aligned}
\max \ & \sum_t \sum_i w_i x_i^t \\
& x_i^t = \sum_{k=1}^m x_{ik}^{ts} = \sum_{k=1}^m x_{ik}^{te}, \forall i, \forall t \in T \\
& x_{ik}^{ts} = 0, \forall t \in T, \forall k \notin S_i \\
& x_{ik}^{te} = 0, \forall t \in T, \forall k \notin E_i \\
& \sum_t \sum_i x_{ik}^{to} \leq 1, \forall k \in M
\end{aligned}$$

$$\sum_t x_{ik}^{ts} \le P_{ik}^s,\ \sum_t x_{ik}^{te} \le P_{ik}^e\ \forall i \in N, \forall k \in M$$
$$0 \le x_i^t, x_{ik}^{ts}, x_{ik}^{te}, x_{ik}^{to},$$

where $x_{ik}^s := \sum_t x_{ik}^{ts} \le P_{ik}^s$ can be interpreted as the probability of i being scheduled and starting at k being at most the marginal probability P_{ik}^s, similar for $x_{ik}^e := \sum_t x_{ik}^{te} \le P_{ik}^e$. For each $k \in M$, we can rewrite $\sum_t \sum_i x_{ik}^{to} \le 1$ as

$$\begin{aligned}
\sum_t \sum_i x_{ik}^{to} &= \sum_t \left(\sum_{i, a_i \le k \le b_i} x_{ik}^{to} + \sum_{i, c_i \le k \le d_i} x_{ik}^{to} + \sum_{i, b_i < k < c_i} x_{ik}^{to} \right) \\
&= \sum_t \left(\sum_{i, a_i \le k \le b_i - 1} \sum_{\ell = a_i}^{k} x_{i\ell}^{ts} + \sum_{i, c_i + 1 \le k \le d_i} \sum_{\ell = k}^{d_i} x_{i\ell}^{te} + \sum_{i, b_i \le k \le c_i} x_i^t \right) \qquad (1) \\
&= \sum_t \sum_{i, b_i \le k \le c_i} x_i^t + \sum_t \left(\sum_{i, a_i \le k \le b_i - 1} \sum_{\ell = a_i}^{k} x_{i\ell}^{ts} + \sum_{i, c_i + 1 \le k \le d_i} \sum_{\ell = k}^{d_i} x_{i\ell}^{te} \right) \le 1,
\end{aligned}$$

where $a_i := \min\{k : k \in S_i\}, b_i := \max\{k : k \in S_i\}, c_i := \min\{k : k \in E_i\}, d_i := \max\{k : k \in E_i\}$. The first equality comes from the fact that the probability i occupying a slot k not in $S_i \cup E_i \cup (b_i, c_i)$ is 0. For the second equality, by definition, any $k \in [b_i, c_i]$ is always occupied by i if it is scheduled, so $x_{ik}^{to} = x_i^t$; for $k \in [a_i, b_i)$, the probability of i being scheduled and occupying k at t is equivalent to the probability of i being scheduled and starting at or before k at t, similar for the ending one. The third equality is derived by rearranging terms. Since $\sum_{i, a_i \le k \le b_i - 1} \sum_{\ell = a_i}^{k} x_{i\ell}^{ts} = \sum_{i, a_i \le k \le b_i - 1} (x_i^t - \sum_{\ell = k+1}^{b_i} x_{i\ell}^{ts})$, and similar for $x_{i\ell}^{te}$, we have

$$\sum_t \left(\sum_{i, a_i \le k \le d_i} x_i^t - \sum_{i, a_i \le k \le b_i} \sum_{\ell = k+1}^{b_i} x_{i\ell}^{ts} - \sum_{i, c_i \le k \le d_i} \sum_{\ell = c_i}^{k-1} x_{i\ell}^{te} \right) \le 1,\ \forall k \in M. \qquad (2)$$

Then we have the following primal-dual pair:

$$\begin{aligned}
\max_{x \ge 0}\ & \sum_t \sum_i w_i x_i^t \\
& x_i^t = \sum_{k=1}^{m} x_{ik}^{ts} = \sum_{k=1}^{m} x_{ik}^{te}, \forall i \in N, \forall t \in T \\
& x_{ik}^{ts} = 0, \forall t \in T, \forall k \notin [a_i, b_i] \\
& x_{ik}^{te} = 0, \forall t \in T, \forall k \notin [c_i, d_i] \\
& \sum_t \left(\sum_{i, a_i \le k \le d_i} x_i^t - \sum_{i, a_i \le k \le b_i} \sum_{\ell = k+1}^{b_i} x_{i\ell}^{ts} - \sum_{i, c_i \le k \le d_i} \sum_{\ell = c_i}^{k-1} x_{i\ell}^{te} \right) \le 1,\ \forall k \in M \\
& \sum_t x_{ik}^{ts} \le P_{ik}^s,\ \sum_t x_{ik}^{te} \le P_{ik}^e, \forall i \in N, \forall k \in M
\end{aligned} \qquad \text{(DLP-P)}$$

$$\begin{aligned}
\min_{\mu, \nu, u \ge 0, y, z}\ & \sum_{r=1}^{m} \mu_r + \sum_{i=1}^{n} \sum_{k \in S_i} P_{ik}^s \nu_{ik}^s + \sum_{i=1}^{n} \sum_{k \in E_i} P_{ik}^e \nu_{ik}^e + \sum_{i=1}^{n} \sum_{t=1}^{T} u_{it} \\
\text{s.t.}\ & y_{it} + z_{it} + u_{it} + \sum_{r=a_i}^{d_i} \mu_r \ge w_i, \forall i \in N,\ \forall t \in T, \\
& -y_{it} - \sum_{r=a_i}^{k-1} \mu_r + \nu_{ik}^s \ge 0, \forall i \in N,\ \forall t \in T,\ \forall k \in S_i, \\
& -z_{it} - \sum_{r=k+1}^{d_i} \mu_r + \nu_{ik}^e \ge 0, \forall i \in N,\ \forall t \in T,\ \forall k \in E_i.
\end{aligned} \qquad \text{(DLP-D)}$$

For (DLP-D), by setting y, z, u to be 0, (DLP-D) is equivalent to,

$$\begin{aligned}
\min_{\mu, u \ge 0}\ & \sum_{r=1}^{m} \mu_r + \sum_{i=1}^{n} \sum_{k \in S_i} P_{ik}^s \sum_{r=a_i}^{k-1} \mu_r + \sum_{i=1}^{n} \sum_{k \in E_i} P_{ik}^e \sum_{r=k+1}^{d_i} \mu_r \\
\text{s.t.}\ & \sum_{r=a_i}^{d_i} \mu_r \ge w_i,\ \forall i \in N.
\end{aligned} \qquad \text{(LP-Dpes)}$$

Consider the pessimistic interval graph G_{pes}, where each interval i is realized as $[a_i, d_i]$, i.e., as wide as possible. Consider (LP-D) defined on G_{pes}, and let μ_{pes} be its optimal solution, $\alpha_{\text{pes}} := \alpha(G_{\text{pes}}) = \sum_r [\mu_{\text{pes}}]_r$ be its optimal value. Since $\mu = \mu_{\text{pes}}, y = z = u = 0$ is feasible to (DLP-D), the optimal value of (DLP-D), α^*, is at most $\alpha_{\text{pes}} + \sum_{i=1}^{n} \sum_{k \in S_i} P_{ik}^s \sum_{r=a_i}^{k-1} [\mu_{\text{pes}}]_r + \sum_{i=1}^{n} \sum_{k \in E_i} P_{ik}^e \sum_{r=k+1}^{d_i} [\mu_{\text{pes}}]_r$, where

$$\sum_{i=1}^{n} \sum_{k \in S_i} P_{ik}^s \sum_{r=a_i}^{k-1} [\mu_{\text{pes}}]_r = \sum_r [\mu_{\text{pes}}]_r \sum_i \sum_{k, r \in S_i, k > r} P_{ik}^s = \sum_r [\mu_{\text{pes}}]_r \sum_{i \text{ s.t. } r \in S_i} (1 - P_{ir}^o),$$

similar for the other term, we have

$$\sum_{i=1}^{n}\sum_{k\in E_i} P^e_{ik}\sum_{r=k+1}^{d_i}[\mu_{\text{pes}}]_r = \sum_r [\mu_{\text{pes}}]_r \sum_{i \text{ s.t. } r\in E_i}(1-P^o_{ir}).$$

Thus,

$$\sum_r [\mu_{\text{pes}}]_r \le \alpha^* \le \sum_r [\mu_{\text{pes}}]_r \left(1+\sum_{i \text{ s.t. } r\in S_i}(1-P^o_{ir})+\sum_{i \text{ s.t. } r\in E_i}(1-P^o_{ir})\right).$$

The bound is tight when the problem is deterministic because for every $r\in M$, $\sum_{i \text{ s.t. } r\in S_i}(1-P^o_{ir}) = \sum_{i \text{ s.t. } r\in E_i}(1-P^o_{ir}) = 0$.

If the length of each task is fixed to be 1, that is, $s_i = e_i$, the problem is trivial.

Proposition 2.3. *If the length of each task is always* 1*, the greedy policy is optimal.*

Proof. We use an adversarial argument. Consider an adversary that places tasks N into $[m]$ positions in advance according to their distributions D^s_i, D^e_i for each $i\in N$. Since the length of each task is 1, if the positions of the tasks are known, it is optimal to pick the largest weighted task of each position. Without loss of generality, assume the weights of the largest weighted task of each position to be $w_1 \ge w_2 \ge \dots w_m \ge 0$. Without knowing the positions, the greedy policy would pick $w_1, w_2, \dots, w_m$ which is equivalent to the case where positions are known. Thus, for any realization, the greedy policy returns the set of non-conflicting tasks with the maximum weights. □

3 Conservative DSRSE

In this section, we consider another version of DSRSE, where a task is deleted whenever it possibly conflicts with the committed tasks.

One is given M, N, w, D like DSRSE. Once s_i, e_i of a committed task are determined, for any remaining task j, instead of flipping a coin, delete j if any slot in its support is occupied by committed tasks; otherwise, keep j. The goal remains to schedule tasks sequentially and maximize the expected total weight of scheduled tasks. We call this problem *conservative dynamic scheduling with random start and end times (CDSRSE).*

Example 3.1. *Consider an instance with $N=[2], M=[3]$, and unit weights, where $D^s_1 = \mathbb{1}_{\{1\}}, D^e_1 = \mathbb{1}_{\{2\}}$, $D^s_2 = U(\{2,3\}), D^e_2 = \mathbb{1}_{\{3\}}$. If the scheduler commits to task* 1 *first, then task* 1 *occupies* $[1,2]$*. For CDSRSE, task* 2 *is deleted for sure since position* 2 *is in its support, which is occupied by task* 1*.*

Even CDSRSE remains NP-hard.

Theorem 3.2 ([8]). *CDSRSE is still NP-hard.*

Proof. Consider the instance with $n+1$ tasks and $2n$ slots. Task 0 takes slots $1,\dots,n$ and task i takes slot i with probability p_i and $n+i$ with $q_i := 1-p_i$. Analogous to the proof in [8], any policy of this problem is determined by when to select task 0, and what is selected before trying to select task 0. Then the problem can be formulated as:

$$\max_{S\subseteq N}\left\{\sum_{i\in S} w_i + (1-\prod_{i\in S} q_i)\sum_{j\notin S} w_j + \prod_{i\in S} q_i w_o\right\}. \tag{Conserv-JS}$$

By rearranging, we get

$$\max_{S\subseteq N}\left\{\sum_{i\in N} w_i + \prod_{i\in S} q_i (w_0 - \sum_{j\notin S} w_j)\right\}.$$

and by dropping constant terms and applying natural logarithm, we get

$$\max_{S\subseteq N}\left\{\sum_{i\in S}\ln q_i + \ln(w_0 - \sum_{j\notin S} w_j)\right\}.$$

Given a set of numbers N with positive weights a_i , $i\in N$, the partitioning problem asks for a subset $S\subseteq N$ such that $\sum_{i\in S} a_i = \sum_{j\notin S} a_j$; without loss of generality, we assume that $\sum_{i\in N} a_i = 2$. By setting parameters of the problem above as $q_i = e^{-a_i}$, $p_i = 1 - e^{-a_i}$, $w_i = a_i$,and $w_0 = \sum_{i \in N} a_i$,

$$\max_{S \subseteq N} \left\{ -\sum_{i \in S} a_i + \ln(\sum_{i \in S} a_i). \right\}$$

Proceeding similarly to [1], a set $S \subseteq N$ with $\sum_{i \in S} a_i = 1$ if and only if optimal solution to the above problem is -1. $\square$

Unlike DSRSE, the distributions D_i^s, D_i^e of CDSRSE are never updated for remaining tasks. Thus D_i^s, D_i^e are independent of the stage t where i is scheduled, and x_{ik}^{ts} can always be expressed as $x_{ik}^{ts} = P_{ik}^s x_i^t$, similarly for x_{ik}^{te}, x_{ik}^{to}. Then, we can express the constraint $\Pr[k \text{ occupied by } i] \leq 1$ as $\sum_{i=1}^n P_{ik}^o x_i \leq 1$. Consider the LP relaxation and its dual:

$$\begin{aligned} \max \quad & \sum_{i=1}^n w_i x_i \\ & \sum_{i=1}^n P_{ik}^o x_i \leq 1 \\ & x \geq 0, \end{aligned} \qquad \text{(CDLP-P)}$$

$$\begin{aligned} \min \quad & \sum_{r=1}^m \mu_r \\ & \sum_{r=1}^m P_{ir}^o \mu_r \geq w_i, \ \forall i \in N \\ & \mu \geq 0. \end{aligned} \qquad \text{(CDLP-D)}$$

Let $p^* := \min\{P_{ir}^o : P_{ir}^o > 0, i \in N, r \in M\}$. Similar to the above, consider the pessimistic interval graph G_{pes} and the corresponding optimal solution μ_{pes} of (LP-D) defined on G_{pes}. Let $\hat{\mu} := \mu_{\text{pes}}/p^*$, then for each $i \in N$,

$$\sum_r P_{ir}^o \hat{\mu}_r \geq \sum_{r \in [a_i, d_i]} [\mu_{\text{pes}}]_r \geq w_i.$$

Thus $\hat{\mu}$ is feasible to (CDLP-D) and $\sum_r \hat{\mu}_r = \frac{1}{p^*} \alpha_{\text{pes}}$. We have $\alpha_{\text{pes}} \leq OPT \leq \frac{1}{p^*} \alpha_{\text{pes}}$.

Remark 3.3. *We emphasize that $\sum_{i=1}^n P_{ik}^o x_i \leq 1$ is not a valid inequality for the original DSRSE because the distribution of start and end times of i depends on the stage at which it is committed. Consider Example 2.2 and a policy that always schedules task 1 first and then task 2 if possible. Under this policy, the probability that slot 2 is occupied is $1 \cdot x_1 = 1$, but $P_{12}^o x_1 + P_{22}^o x_2 = 1 \cdot x_1 + \frac{1}{2} x_2 = 1 + \frac{1}{4} > 1$. For Example 3.1, after task 1 is scheduled, the probability of scheduling task 2 is 0, so $P_{12}^o x_1 + P_{22}^o x_2 = 1$.*

4 Uniform Weights

In this section, we consider the case where the weights of tasks are uniform. Consider the case where the weights of all tasks are uniform. Without loss of generality, let $w_i = 1$ for every task, and α_{pes} be the optimal value of the pessimistic interval graph. For G_{pes}, because of the uniform weights, there is a set $\mathscr{J} \subseteq M$ with $|\mathscr{J}| = \alpha_{\text{pes}}$ such that each task covers exactly one element in $\mathscr{J}$, which is the minimum clique cover derived from (LP-D). In particular, for each task i, there exists exactly one $k \in \mathscr{J}$ such that $k \in [a_i, d_i]$. Let x be an optimal solution of (DLP-P), and α be the optimal value. By (2), we have for each $k \in \mathscr{J}$,

$$\begin{aligned} \alpha = \sum_i x_i = \sum_t \sum_i x_i^t &\leq 1 + \sum_t \left(\sum_{i, a_i \leq k \leq b_i} \sum_{\ell=k+1}^{b_i} x_{i\ell}^{ts} + \sum_{i, c_i \leq k \leq d_i} \sum_{\ell=c_i}^{k-1} x_{i\ell}^{te} \right) + \sum_t \sum_{i, P_{ik}^o = 0} x_i^t \\ &= 1 + \sum_{i, a_i \leq k \leq b_i} \sum_{\ell=k+1}^{b_i} x_{i\ell}^s + \sum_{i, c_i \leq k \leq d_i} \sum_{\ell=c_i}^{k-1} x_{i\ell}^e + \sum_{i, P_{ik}^o = 0} x_i. \end{aligned}$$

Let k_i be the unique element in $\mathscr{J}$ covering $i \in [n]$. Sum over both sides over $\mathscr{J}$, we get

$$\begin{aligned} \alpha * \alpha_{\text{pes}} &\leq \alpha_{\text{pes}} + \sum_{k \in \mathscr{J}} \left(\sum_{i, a_i \leq k \leq b_i} \sum_{\ell=k+1}^{b_i} x_{i\ell}^s + \sum_{i, c_i \leq k \leq d_i} \sum_{\ell=c_i}^{k-1} x_{i\ell}^e \right) + (\alpha_{\text{pes}} - 1) \sum_i x_i \\ \iff \alpha &\leq \alpha_{\text{pes}} + \sum_{k \in \mathscr{J}} \left(\sum_{i, a_i \leq k \leq b_i} \sum_{\ell=k+1}^{b_i} x_{i\ell}^s + \sum_{i, c_i \leq k \leq d_i} \sum_{\ell=c_i}^{k-1} x_{i\ell}^e \right) \\ &= \alpha_{\text{pes}} + \sum_i \sum_{\ell=k_i+1}^{b_i} x_{i\ell}^s + \sum_{\ell=c_i}^{k_i-1} x_{i\ell}^e \end{aligned}$$

$$\leq \alpha_{\text{pes}} + \sum_i (1 - P^o_{ik_i})$$

where the first and third lines are from the fact that each i is not covered by exactly $\mathscr{J} \setminus \{k_i\}$ whose cardinality is $\alpha_{\text{pes}} - 1$. And the last line is from the constraint $\sum_t x^{ts}_{i\ell} \leq P^s_{i\ell}$, same for $x^e_{i\ell}$.

For CDSRSE, we have $x^s_{i\ell} = x_i P^s_{i\ell}$ and $x^e_{i\ell} = x_i P^e_{i\ell}$. Thus we have,

$$\alpha \leq \alpha_{\text{pes}} + \sum_i \sum_{\ell=k_i+1}^{b[i]} x^s_{i\ell} + \sum_{\ell=c[i]}^{k_i-1} x^e_{i\ell} = \alpha_{\text{pes}} + \sum_i (1 - P^o_{ik_i}) x_i \leq \alpha_{\text{pes}} + \alpha * (1 - P^o_{i^*k[i^*]}),$$

where $P^o_{i^*k[i^*]} = \min_i \{P^o_{ik_i}\} > 0$, so $\alpha \leq \frac{1}{P^o_{i^*k[i^*]}} \alpha_{\text{pes}}$.

5 Semidefinite Program Relaxations

In this section, we consider lifting the variables to the *positive semidefinite (PSD) cone*. Consider an optimal policy π^*, which returns the scheduling $x^{(\ell)} \in \{0,1\}^{(T\times N)}$ for the realized interval graph $G^{(\ell)}$. Then $x^{(\ell)}_{it} = 1$ if task i is scheduled at stage t and 0 otherwise. We lift it to

$$\begin{bmatrix} 1 \\ x^{(\ell)} \end{bmatrix} \begin{bmatrix} 1 & {x^{(\ell)}}^\top \end{bmatrix} = \begin{bmatrix} 1 & {x^{(\ell)}}^\top \\ x^{(\ell)} & x^{(\ell)} {x^{(\ell)}}^\top \end{bmatrix}.$$

Let $p^{(\ell)}$ be the probability of the random interval graph being realized as $G^{(\ell)}$, we have

$$\begin{bmatrix} 1 & {x^*}^\top \\ x^* & X^* \end{bmatrix} := \sum_\ell p^{(\ell)} \begin{bmatrix} 1 & {x^{(\ell)}}^\top \\ x^{(\ell)} & x^{(\ell)} {x^{(\ell)}}^\top \end{bmatrix},$$

so $w^\top x^* = \text{OPT}_{\text{DP}}$. We consider relaxations which combines the constraints in (DLP-P) and relaxes the above matrix to the PSD cone,

$$\max \sum_{t\in T} \sum_{i\in N} w_i x^t_i \quad \text{s.t.} \begin{cases} x^t_i = \sum_k x^{ts}_{ik} = \sum_k x^{te}_{ik}, \forall (i,t) \in N \times T, \\ x^{ts}_{ik} = 0, k \notin [a_i, b_i], \\ x^{te}_{ik} = 0, k \notin [c_i, d_i], \\ \sum_t \left(\sum_{i, a_i \leq k \leq d_i} x^t_i - \sum_{i, a_i \leq k \leq b_i} \sum_{\ell=k+1}^{b_i} x^{ts}_{i\ell} - \sum_{i, c_i \leq k \leq d_i} \sum_{\ell=c_i}^{k-1} x^{te}_{i\ell} \right) \leq 1, \forall k \in M, \\ \sum_t x^{ts}_{ik} \leq P^s_{ik},\ \sum_t x^{te}_{ik} \leq P^e_{ik}, \forall (i,k) \in N \times M, \\ \begin{pmatrix} 1 & x^\top \\ x & X \end{pmatrix} \succeq 0,\ X \geq 0,\ \text{Diag}(X) = x, \\ X_{(i,t),(j,t)} = 0\ (i \neq j),\ \ X_{(i,t),(i,\tau)} = 0\ (t \neq \tau), \\ x^t_i = \sum_j X_{(i,t),(j,t-1)}, t \geq 2, \end{cases} \tag{SDP-P}$$

For $X_{(i,t),(j,\tau)}$, one can understand it as the probability of scheduling i at t and j at τ. In addition to the LP relaxations (DLP-P), more structures are added. $X_{(i,t),(j,t)} = 0, \forall t \in T, i \neq j$ can be interpreted as no two tasks being scheduled at the same stage and $X_{(i,t),(i,\tau)} = 0, \forall i \in N, t \neq \tau$ means that one task cannot be scheduled at two different stages. The last inequality $x^t_i = \sum_j X_{(i,t),(j,t-1)}$ means that the probability of scheduling i at stage t is the probability of scheduling some task at stage $t-1$ and i at t.

By definition, this gives a tighter upper bound for DSRSE than (DLP-P).

6 Dynamic Heuristics and Computational Experiments

We conducted all of our computational experiments on a laptop computer with an Apple M1 Pro CPU and 16 GB of RAM, solving SDPs using COPT [4] and performing other computations using Julia.

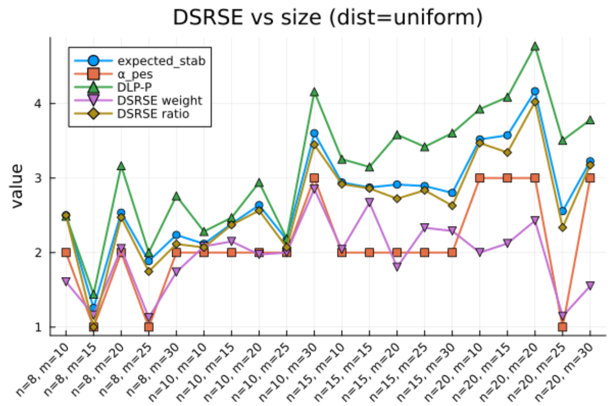
In this section, we test the bounds derived above. For each task i, we first determine the maximum support size m_s by sampling from $\text{Unif}(1,m)$, then uniformly generate a_i from $\text{Unif}(1,m)$, set $b_i := a_i + \text{Unif}(0, m_s - 1)$, and similarly generate c_i from b_i and d_i from c_i. We try uniform, Gaussian, and exponential distributions for the start and end times by truncating them onto the supports.

Since computing the optimal values of DSRSE and CSDRSE is impractical, we simulate the expected maximum scheduling $\mathbb{E}[\alpha(G)]$ by sampling the start and end times of each task and solving the maximum independent set problem, which is an upper bound for the optimal values of DSRSE and CDSRSE. We compare it with the optimal value of (DLP-P), (CDLP-P), $\alpha(G_{\text{pes}})$, α_{pes}/p^*, (SDP-P) and $\sum_r [\mu_{\text{pes}}]_r \left(1+\sum_{i \text{ s.t. } r\in S_i}(1-P^o_{ir})+\sum_{i \text{ s.t. } r\in E_i}(1-P^o_{ir})\right)$. All the weights are i.i.d. $\text{Unif}(0,1)$.

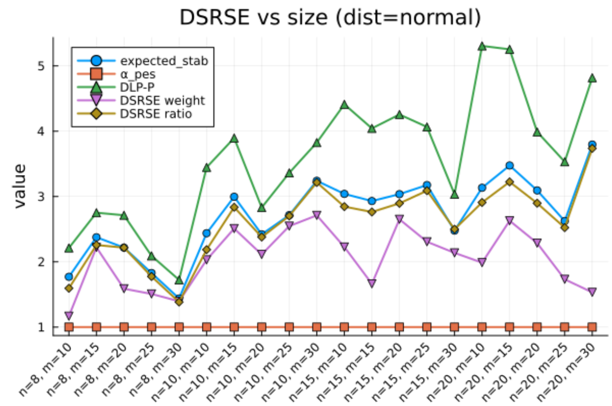
For (CDLP-D), there are many feasible dual solutions which produce bounds. Consider $\mu_r := \max_i\{w_i/\sum_{k=1}^m P^o_{ik} : P^o_{ir} > 0\}$, then

$$\sum_{r=1}^m P^o_{ir}\mu_r \geq \sum_{r=1}^m P^o_{ir}\frac{w_i}{\sum_{k=1}^m P^o_{ik}} = w_i,$$

we also consider the heuristic of picking tasks with the largest $w_i/\mathbb{E}[\text{length of } i] = w_i/\sum_r P^o_{ir}$, which can be interpreted as the weight gained per expected occupied position.



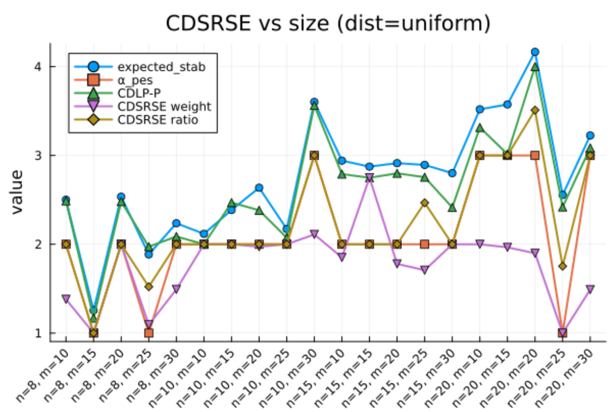
(a) DSRSE with uniform start and end times.



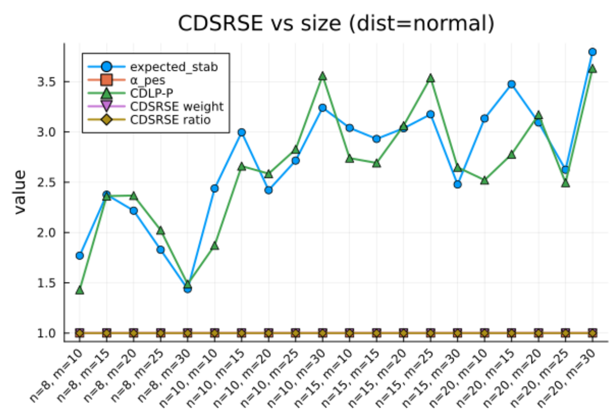
(b) DSRSE with Gaussian start and end times.

Figure 1: DSRSE

From Figures 1a and 1b, LP-DSRSE denotes the optimal value of (DLP-P), DSRSE weight is the expected value of the greedy policy that schedules the largest-weight task, and DSRSE ratio is the expected value of the greedy policy that schedules the largest $w_i/\mathbb{E}[\text{length of } i]$. For both distributions, DSRSE ratio is close to the expected stability number. The optimal solution is lower bounded by DSRSE ratio and upper bounded by the expected stability number, so DSRSE ratio is a practical heuristic. By contrast, (DLP-P) does not give a useful bound in this case.



(a) CDSRSE with uniform start and end times.



(b) CDSRSE with Gaussian start and end times.

Figure 2: CDSRSE

For CDSRSE, the greedy policies do not perform as well because conflicts are enforced conservatively. Even though the LP relaxation (CDLP-P) provides bounds close to the expected stability number, we do not know its actual performance since both are upper bounds for CDSRSE.

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